

MAHBUB RAHIM

FULL STACK ENGINEER
mahbub.practicecompanions.com

PROFILE

Experienced full-stack web engineer with AI-driven solution experience. Skilled in JavaScript, Python, browser APIs, and popular frameworks such as React, Next.js, and the MERN stack.

Recently founded and developed PracticeCompanions, an AI-powered evaluation system for Duolingo and IELTS practice, strengthening my interest in machine learning and NLP.

EXPERIENCE

PracticeCompanions (Founder & Full-Stack Engineer)

August 2023- Present

AI POWERED PRACTICE PLATFORM FOR DUOLINGO & IELTS

- Designed and implemented core features including dynamic quiz modules, mock tests, instant evaluation with AI feedback, and detailed progress tracking.
- Developed both frontend and backend architecture using React, Next.js, Node.js, Express, and PostgreSQL, ensuring a responsive, user-friendly experience.
- Integrated GPT-4 API (fine-tuned) for real-time scoring and personalized feedback.
- Deployed scalable infrastructure using AWS Lambda, Docker, and CI/CD pipelines to support high user traffic.
- Managed end-to-end operations: coding, design, database design, cloud deployment, content strategy, SEO, and analytics tracking with Google Analytics.
- Tech Stack: React, Next.js, Node.js, Express, PostgreSQL, Redux, JWT, AWS Lambda, Docker, Google Analytics, OpenAI GPT-4 API

FULL STACK WEB DEVELOPER | APRIL 2022 -MAY 2024

MEGH TECHNOLOGIES LTD.

- Built and delivered production-grade web applications for diverse clients using React, TypeScript, Laravel, Next.js, Express.js and Vue.
- Integrated Google Analytics to track and analyze user behavior for improved product insights and decision-making.
- Increased client acquisition by 29% by implementing i18n support across applications.
- Developed fully responsive custom React data tables optimized for all device types.
- Performed unit and end-to-end testing using Jest, Mocha, React Testing Library, and Cypress to ensure application quality and reliability.
- Conducted peer code reviews and mentored team members to improve code quality and development practices.

TEACHING ASSISTANT | MAY - AUGUST 2022

AMERICAN INTERNATIONAL UNIVERSITY- BANGLADESH

- Assisted Professor during the lab.
- Solved problem while students faced problem during lab class.
- Checked students assignments.

EDUCATION

AMERICAN INTERNATIONAL UNIVERSITY - BANGLADESH | 2017- 2022

Bachelor of Science in Computer Science and Engineering

- GPA 3.19 out Of 4
- Relevant coursework: Data Structures, Algorithms, Artificial Intelligence and Expert System, Web Technologies, Software Engineering.

JAHANGIRPUR GOVT. COLLEGE, NAOGAON - BANGLADESH | 2016

Higher Secondary Certificate

- GPA 4.58 out Of 5

PUBLICATION

UNDERGRADUATE THESIS: ENERGY OPTIMIZATION IN IOT. (UNDER REVIEW)

Achieved 90% energy optimization in IoT using PCA-based compression and Wavelet Difference Reduction, minimizing memory/computation while ensuring reversible data integrity.

SKILLS

- | | |
|---------------|----------------|
| • Javascript | • Fast API |
| • Typescript | • Tailwind CSS |
| • React | • AWS |
| • NextJS | • Docker |
| • ReactNative | • Kubernetes |
| • Python | • Mocha |
| • Django | • Postgres |

REFERENCES

DR. MOHAMMAD ABDUL MANNAN

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